

Commodity Price Shocks as an Early-Warning Signal for US Initial Jobless Claims

Milestone 02 manuscript draft: Section II (Data and Preprocessing), Section III (Exploratory Data Analysis) and Section IV (Hypotheses and Analytical Plan)

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Abstract

Official unemployment statistics arrive weeks after layoffs occur. This project asks whether sudden movements in the prices of four physical input commodities (crude oil, natural gas, copper, iron ore) anticipate increases in US weekly initial jobless claims, and with what lag. This draft documents the construction of a weekly master dataset of 1,500 observations and 17 variables (1998-01-03 to 2026-09-26) from the FRED API and Yahoo Finance, the alignment of daily and monthly series onto the Department of Labor's claims week, the missing-data and transformation decisions, and an exploratory analysis. Claims are heavily right-skewed (skewness 11.2) and the 2020 pandemic weeks inflate the standard deviation threefold, which masks every other recession under a z-score outlier rule. Week-over-week commodity shocks are only weakly cross-correlated with claims changes ($|r| \leq 0.13$), but the lag profile is informative: oil and copper co-move with claims at lags 0–1 with a negative sign, and a top-decile natural-gas move four weeks earlier doubles the probability of a claims spike (8.9 % to 18.5 %, $\chi^2 = 10.8$, $p = 0.001$). Five formal hypotheses and their variable mapping close the draft.

Research question. Do price shocks and volatility in core energy and industrial-metal commodities predict increases in US initial jobless claims, and what is the delay between shock and claims response?

II. Data and Preprocessing

II.1 Data sources and collection

All raw series are collected programmatically by `src/collect.py` and cached with a provenance manifest (Table 1). Six series come from the Federal Reserve Economic Data (FRED) keyless CSV endpoint (<https://fred.stlouisfed.org/graph/fredgraph.csv?id=SERIES>): weekly initial claims (ICSA), daily WTI crude oil (DCOILWTICO), daily Henry Hub natural gas (DHHNGSP), the monthly NBER recession indicator (USREC) and the two monthly metal benchmarks (PCOPPUSDM, PIORECRUSDM) that are kept for reference only. FRED encodes non-trading days as '.', which is parsed to a missing value at load time. Following the Milestone 01 source-selection test, daily copper and iron ore prices come from Yahoo Finance via the `yfinance` package: COMEX copper front-month futures (HG=F, USD per pound) and iron ore 62 % Fe CFR China futures (TIO=F, USD per tonne). A direct query of the Yahoo chart endpoint for 2007–2010 returns "Data doesn't exist" for TIO=F: the contract was listed on 14 October 2010, and HG=F history begins on 30 August 2000. Both gaps are therefore structural.

Table 1. Raw series collected, with row counts and coverage as recorded in `data/raw/manifest.json` (pulled 2026-10-04 10:48 UTC).

Series	Description	Freq.	Rows	First	Last
ICSA	Initial Claims (weekly, SA, persons)	weekly	1500	1998-01-03	2026-09-26
DCOILWTICO	WTI crude oil spot (daily, USD/bbl)	daily	7499	1998-01-01	2026-09-29
DHHNGSP	Henry Hub natural gas spot (daily, USD/MMBtu)	daily	7499	1998-01-01	2026-09-29

USREC	NBER recession indicator (monthly, 0/1)	monthly	345	1998-01-01	2026-09-01
PCOPPU\$DM	Copper global price (monthly, USD/tonne) -- reference	monthly	343	1998-01-01	2026-07-01
PIORECR\$DM	Iron ore global price (monthly, USD/tonne) -- reference	monthly	343	1998-01-01	2026-07-01
HG_F	COMEX copper futures (daily, USD/lb)	daily	6552	2000-08-30	2026-10-02
TIO_F	Iron ore 62% Fe CFR China futures (daily, USD/tonne)	daily	4003	2010-10-14	2026-10-02

II.2 Alignment and the master dataset

The observational unit is one calendar week ending Saturday, the Department of Labor's reference week for initial claims. The claims series is never resampled; every other series is brought onto its grid (src/build_master.py). Daily prices are collapsed into Sunday–Saturday bins and summarised by the weekly mean (the base weekly price) and a trailing 20-trading-day volatility of daily log returns sampled on the last day of the week. The monthly recession flag is forward-filled to weeks. All weekly frames are left-joined onto the claims index, so no commodity week without a claims observation enters the table. Per commodity we then derive the week-over-week percentage change of the weekly mean (the price shock). Lags 1–16 of the shock, the proposal's exploratory window, are generated on demand by add_lags() rather than stored, which keeps the master at 17 columns. Copper is converted from USD per pound to USD per tonne ($\times 2,204.62$) so that it is comparable with the FRED benchmark.

The weekly mean, rather than the close, is the level feature because it reflects what a business paying for inputs over the week actually faced. The week ending 25 April 2020 illustrates the difference: it closes at 15.99 USD/bbl but its mean is 3.32 USD/bbl because of the −37.63 USD/bbl WTI settlement on 20 April. That single negative print has an undefined log return and is excluded from the return-based volatility features only.

Table 2. Master analytical dataset at a glance.

Item	Value
Observations (rows)	1,500 weeks
Variables (columns)	17
Observational unit	1 row = 1 calendar week ending Saturday (the US Department of Labor's 'week ending' convention for initial claims)
Time span	1998-01-03 to 2026-09-26
Target variable	icsa (weekly initial claims, persons) and its derived log / % change / spike indicator
Commodity predictors	4 commodities $\times$ (base weekly mean price, weekly % price shock, 4-week volatility); lags 1–16 of the shock are created on demand by add_lags()
Controls / dummies	NBER recession (usrec), COVID period (covid_period)
Raw sources	FRED API (6 series), Yahoo Finance via yfinance (2 futures tickers)

II.3 Missing data

Table 3. Missing values in every column of the master dataset (17 columns, 1,500 weeks).

Variable	Missing (n)	Missing (%)
icsa	0	0.00
log_icsa	0	0.00
icsa_pct_chg_1w	1	0.07
oil_mean	4	0.27
oil_pct_1w	8	0.53
oil_vol_4w	6	0.40
gas_mean	121	8.07
gas_pct_1w	203	13.53
gas_vol_4w	126	8.40
copper_mean	139	9.27
copper_pct_1w	140	9.33
copper_vol_4w	141	9.40
iron_mean	667	44.47
iron_pct_1w	668	44.53
iron_vol_4w	669	44.60
usrec	0	0.00
covid_period	0	0.00

Table 4. Share of weeks (%) with a missing weekly mean price, by year, 1998–2011 (no missing weeks after 2011).

commodity	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
oil	2	6	0	0	0	0	0	0	0	0	0	0	0	0
gas	17	25	13	12	29	37	35	21	31	13	0	0	0	0
copper	100	100	66	0	0	0	0	0	0	0	0	0	0	0
iron	100	100	100	100	100	100	100	100	100	100	100	100	79	0

The target is complete. Oil is missing in only 4 weeks (1998–99). Natural gas is missing in 121 weeks (8.1 %), all before April 2007, because FRED's daily Henry Hub series is intermittent in its early years. Copper (9.3 %) and iron ore (44.5 %) are missing for every week before their futures histories begin (September 2000 and October 2010) and complete afterwards.

Handling rationale. No value is imputed. Mean or median imputation of a price level that did not exist would inject a constant into a time series and destroy the week-to-week variation the project studies; interpolating across a ten-year hole would be fabrication. Back-filling iron ore from the FRED monthly benchmark was considered and rejected: before 2010 iron ore was priced on annual contracts, so the FRED series is a flat step function (36.63 USD/t throughout 2007) with no weekly information, and mixing two sources in one column contradicts the single-source decision of Milestone 01. Because the target is complete, no row is dropped from the master table; each model will instead apply listwise

deletion on the columns it uses, so a copper model trains on 1,361 weeks (2000–2026) and an iron-ore model on 833 weeks (2010–2026).

Selection bias. Listwise deletion is innocuous only when missingness is unrelated to the outcome. Here it is related to time, and time to the outcome: any specification that includes iron ore observes neither the 2001 recession nor the 2008–09 financial crisis, the two largest claims episodes other than COVID-19, and will be estimated on a sample with a narrower range of labour-market stress. Metal-based models will therefore be reported both with and without iron ore. The pre-2007 gas gaps are scattered across ordinary years and are closer to missing at random.

II.4 Transformations and encoding

Table 5. Effect of the natural-log transform on skewness.

Variable	Skewness (raw)	Skewness (log)
icsa	11.22	2.30
oil_mean	0.11	-0.90
gas_mean	1.75	0.55
copper_mean	0.01	-1.08
iron_mean	0.47	-0.34

Log transformation. Initial claims are extremely right-skewed (11.2): only 50 of 1,500 weeks exceed 700,000, but the pandemic weeks reach 6.14 million. $\log(\text{ICSA})$ reduces skewness to 2.3 and is the response scale we intend to model; a unit change in log claims is a constant percentage change, which is also how layoff waves are discussed by policymakers. Natural gas is the only commodity with material right skew (1.75), reduced to 0.55 by the log. Oil, copper and iron ore are close to symmetric in levels and the log over-corrects oil and copper into moderate left skew, so their raw levels remain the primary features.

Scaling. Standardisation (z-score) and Min-Max normalisation of claims, log claims and the four weekly mean prices are computed and reported in the notebook but not stored in the master. Standardisation makes coefficients comparable across commodities measured in dollars per barrel, per MMBtu and per tonne; Min-Max serves bounded-input models. Because the statistics are full-sample, storing scaled columns would bake look-ahead information into the file; in Milestone 03 the scalars are fitted on the training window inside the model pipeline.

Table 6. Boolean dummy indicators in the master dataset.

Indicator	Weeks = 1	Share
usrec	120	0.08
covid_period	43	0.03

Categorical encoding. Two nominal string variables, calendar quarter (Q1–Q4) and regime (expansion/recession from the NBER indicator), are converted with `pandas.get_dummies(drop_first=True)` into 0/1 flags, with Q1 and expansion as reference levels to avoid the dummy-variable trap. Only the regime flag is stored, as `usrec`; the quarter dummies are shown for the mechanics but not kept because initial claims are already seasonally adjusted. The other stored flag is `covid_period` (March–December 2020). A target-derived `icsa_spike` indicator, marking the top decile of weekly claims increases (rises above 5.2 %), is created inside the notebook for the exploratory analysis. Only 120 weeks (8 %) fall in a recession, which already warns that regime-interaction hypotheses rest on few observations.

III. Exploratory Data Analysis

III.1 Univariate summary statistics

Table 7. Central tendency and spread of the key numerical variables (weekly observations; n varies with data availability).

Variable	Unit	n	Mean	Median	SD	IQR	Min	Max	Skew
icsa	persons/wk	1500	361,644.67	315,000	330,652.02	146,000	189,000	6,137,000	11.22
icsa_pct_chg_1w	%	1499	0.59	-0.31	25.56	5.15	-20.66	967.40	36.20
oil_mean	USD/barrel	1496	61.12	61.34	26.92	41.17	3.32	142.52	0.11
gas_mean	USD/MMBtu	1379	4.06	3.30	2.15	2.14	1.40	14.82	1.75
copper_mean	USD/tonne	1361	6,462.30	6,763.77	2,823.63	3,501.82	1,359.15	14,787.71	0.01
iron_mean	USD/tonne	833	106.81	104.72	36.17	50.62	38.73	218.40	0.47
oil_pct_1w	%	1492	0.42	0.44	10.99	5.18	-83.48	372.68	25.82
gas_pct_1w	%	1297	0.79	-0.09	14.22	8.49	-76.54	300.57	9.18
copper_pct_1w	%	1360	0.19	0.16	2.89	3.21	-15.95	10.40	-0.30
iron_pct_1w	%	832	0.02	0.08	3.59	2.62	-19.63	17.83	-0.50
oil_vol_4w	log-ret SD	1494	0.02	0.02	0.02	0.01	0.01	0.18	4.29
gas_vol_4w	log-ret SD	1374	0.06	0.04	0.05	0.04	0.01	0.47	3.82
copper_vol_4w	log-ret SD	1359	0.02	0.01	0.01	0.01	0.01	0.07	2.49
iron_vol_4w	log-ret SD	831	0.01	0.01	0.01	0.01	0.00	0.05	1.37

Table 8. Resistant versus non-resistant measures of weekly initial claims with and without the 43 pandemic weeks.

Statistic	All weeks	Excl. Mar–Dec 2020	Change (%)
n	1,500	1,457	-2.90
mean	361,644.67	325,163.35	-10.10
median	315,000	313,000	-0.60
SD	330,652.02	100,782.95	-69.50
IQR	146,000	139,000	-4.80

Mean weekly claims (361,645) sit 15 % above the median (315,000) and the SD (330,652) is 2.3 times the IQR (146,000), the signature of a heavy right tail. Removing the 43 pandemic weeks changes the mean by -10 % and the SD by -70 % but the median by only -0.6 % and the IQR by -5 %: median and IQR are resistant, mean and SD are not, so medians and IQRs are the headline descriptives and the log scale is used in models. Crude oil averages 61 USD/bbl with an almost symmetric distribution; natural gas (mean 4.06 vs median 3.30 USD/MMBtu) is right-skewed. Weekly percentage changes are centred near zero but their tails differ sharply: oil and gas record single-week moves of -83 % / +373 % and -77 % / +301 %, whereas copper and iron ore never move more than ± 20 % in a week.

III.2 Outlier diagnostic

Table 9. Outlier counts under the z-score rule ($|z| > 3$) and the quartile rule (beyond $Q3 + 1.5 \cdot IQR$ or $Q1 - 1.5 \cdot IQR$), for the full sample and excluding March–December 2020.

Variable	z-rule $ z >3$ (count) (full)	z upper bound (full)	IQR-rule (count) (full)	IQR upper bound (full)	z-rule $ z >3$ (count) (excl. 2020)	z upper bound (excl. 2020)	IQR-rule (count) (excl. 2020)	IQR upper bound (excl. 2020)
icsa	18	1,353,600.7	71	606,000	24	627,512.2	37	587,500
icsa_pct_chg_1w	2	77.3	51	10.2	18	13.1	42	10.0
oil_pct_1w	3	33.4	55	10.6	17	13.4	43	10.6
gas_pct_1w	14	43.4	78	17.0	13	43.9	77	16.8
copper_pct_1w	17	8.9	52	6.7	15	8.9	50	6.6
iron_pct_1w	17	10.8	88	5.4	18	10.8	86	5.3

The two rules disagree by an order of magnitude: on weekly claims the z-score rule flags 18 weeks and the IQR rule 71; on the weekly percentage change 2 against 51. The cause is variance inflation. The pandemic weeks lift the SD of claims from 100,783 to 330,652, so the z-rule's upper bound climbs to 1.35 million and everything short of the COVID spike is hidden behind it. This is the masking effect: the most extreme outliers inflate the scale used to detect outliers, concealing the 2008–09 peak (665,000) and every other recession week. The IQR rule's bound (606,000) depends only on the middle half of the data and isolates exactly the recession periods. Re-running the z-rule without the 43 pandemic weeks raises its count to 24, confirming that those weeks had been masking others. The extreme weeks are real events, not errors, and are retained; their influence is managed by the log transform of the target, the covid_period dummy and, where appropriate, robust loss functions.

III.3 Bivariate and multivariate exploration

Table 10. Strongest cross-correlation between the weekly % change in each commodity's price k weeks earlier and the weekly % change in claims, lags 0–16.

Commodity	Strongest lag	r at that lag	n	$\pm 2/\sqrt{n}$
oil	1	-0.09	1492	0.05
gas	3	-0.03	1297	0.06
copper	0	-0.10	1360	0.05
iron	6	-0.13	832	0.07

Table 11. Contingency analysis: probability of a claims spike (top-decile weekly rise) in weeks following a top-decile absolute commodity price move four weeks earlier.

Commodity	Spike rate after shock	Spike rate otherwise	Relative risk	χ^2	p
WTI crude oil	0.129	0.097	1.333	1.197	0.274
Henry Hub natural gas	0.185	0.089	2.064	10.836	0.001
Copper (COMEX)	0.088	0.102	0.868	0.118	0.732
Iron ore 62% Fe	0.143	0.091	1.563	1.738	0.187

Table 12. Group-by comparison of expansion (n = 1,380) and NBER recession (n = 120) weeks.

Regime	icsa (median)	icsa (mean)	icsa_pct_chg_1w (mean)	icsa_spike (mean)	oil_pct_1w (mean)	copper_pct_1w (mean)	oil_vol_4w (mean)
expansion	309,000	335,130.435	-0.187	0.096	0.611	0.246	0.023
recession	432,500	666,558.333	9.463	0.142	-1.705	-0.377	0.041

Levels. In log levels, claims correlate negatively with oil (-0.20) and copper (-0.28) and positively with iron ore (+0.29) and the recession flag (+0.34). The copper result is the classic "Dr. Copper" pattern: copper is bought when factories expand, so high copper prices coincide with low unemployment; oil shows the same demand-side sign. The iron-ore sign is largely a sample artefact of its 2010 start. These level correlations are dominated by shared slow trends and are not evidence about shocks; they mainly show that levels must enter models detrended or as controls, and that oil and copper levels ($r = 0.82$) should not enter a linear model together.

Shocks and lags. Week-over-week changes are only weakly cross-correlated with claims changes (all $|r| < 0.14$), but the lag profile is informative (Figure 3). Oil is most negative at lags 0–1 (an oil price fall coincides with claims rising, as in 2008 and 2020 when collapsing demand drove both), copper peaks at lag 0 and again at lags 7–8, and iron ore has an isolated -0.13 at lag 6. Oil at lags 0–1, copper at lags 0 and 7 and iron ore at lag 6 clear the $\pm 2/\sqrt{n}$ band; no natural-gas lag does. None of this is yet causal, but it tells the modelling stage where to look and warns that the sign may be negative: commodity collapses, not only spikes, precede layoff waves.

Regimes and contingency. Recession weeks have a median claims level of 432,500 against 309,000 in expansions, a mean weekly claims change of +9.5 % versus -0.2 %, and 4-week oil volatility about 1.7 times higher. Conditioning on a top-decile absolute move four weeks earlier, a natural-gas shock is followed by a claims spike in 18.5 % of weeks versus 8.9 % otherwise (relative risk 2.1, $\chi^2 = 10.8$, $p = 0.001$); oil, copper and iron ore shocks do not shift the spike rate significantly at this lag. The gas result is the strongest single bivariate signal in the data and motivates hypothesis H2.

III.4 Figures

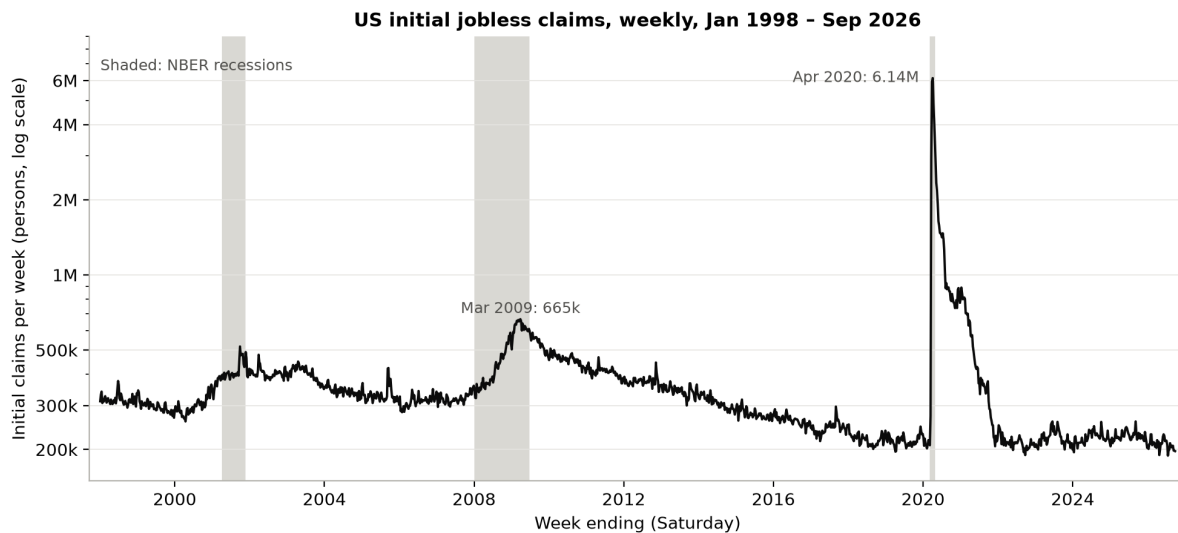


Figure 1. Weekly US initial unemployment-insurance claims (seasonally adjusted, persons per week) on a logarithmic axis, 3 Jan 1998 to 26 Sep 2026. Gray bands mark NBER recessions (2001, 2007–09, 2020). The log scale keeps the 2001 and 2008–09 episodes visible against the 2020 spike. Source: FRED series ICSA.

Weekly mean commodity prices, Jan 1998 – Sep 2026 (gray bands: NBER recessions)

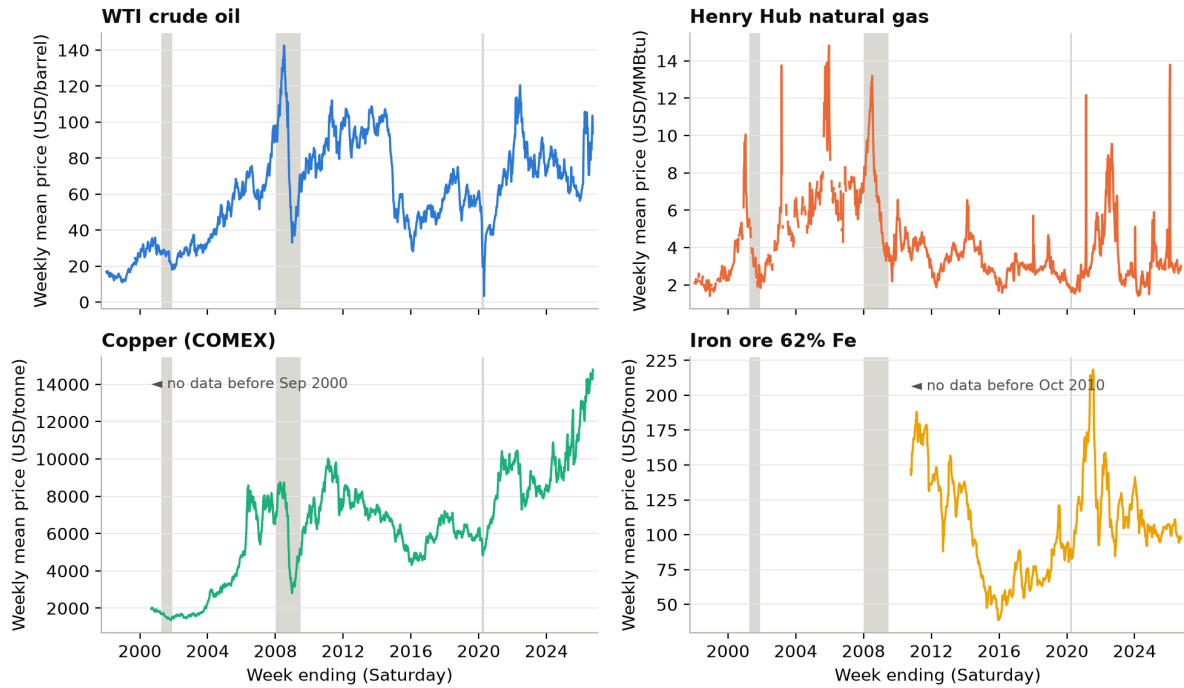


Figure 2. Weekly mean prices of the four input commodities, each on its own axis and physical unit: WTI crude oil (USD/barrel, FRED), Henry Hub natural gas (USD/MMBtu, FRED), COMEX copper futures converted to USD/tonne (Yahoo Finance HG=F) and iron ore 62 % Fe futures (USD/tonne, Yahoo Finance TIO=F). Copper and iron ore begin when their futures histories begin; gray bands mark NBER recessions. All four fall sharply inside the 2008–09 and 2020 recessions.

Cross-correlation between weekly % change in commodity price at week $t-k$ and % change in initial claims at week t (gray band: $\pm 2/\sqrt{n}$, the approximate 95 % interval for zero correlation)

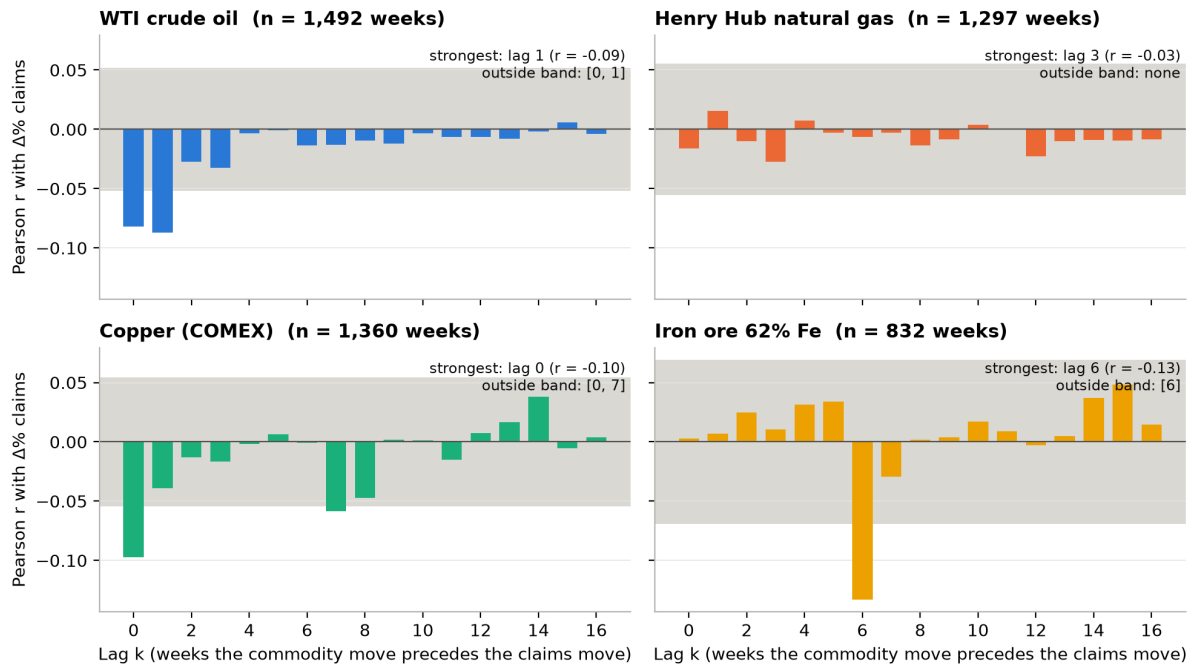


Figure 3. Lead–lag structure between commodity shocks and claims: Pearson correlation between the week-over-week % change in a commodity's weekly mean price k weeks earlier and the week-over-week % change in initial claims, $k = 0-16$; the gray band is $\pm 2/\sqrt{n}$. Oil's association is concentrated at lags 0–1 with a negative sign, copper shows a secondary dip at lags 7–8 and iron ore's one coefficient outside the band is at lag 6.

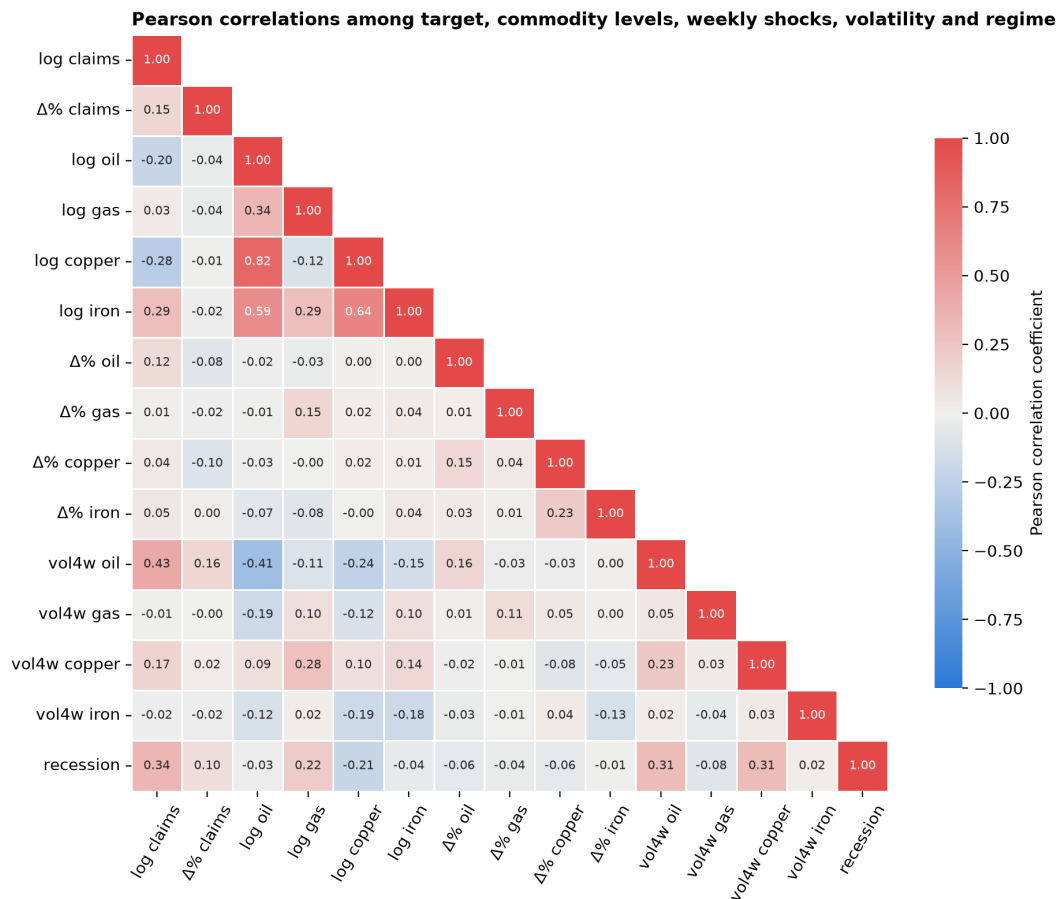


Figure 4. Lower triangle of the Pearson correlation matrix for the target (log level and weekly % change), commodity log levels, weekly % changes, trailing 4-week volatilities and the NBER recession flag (pairwise-complete observations). Diverging scale: blue negative, gray zero, red positive. Levels are strongly inter-correlated; the % change block is nearly orthogonal.

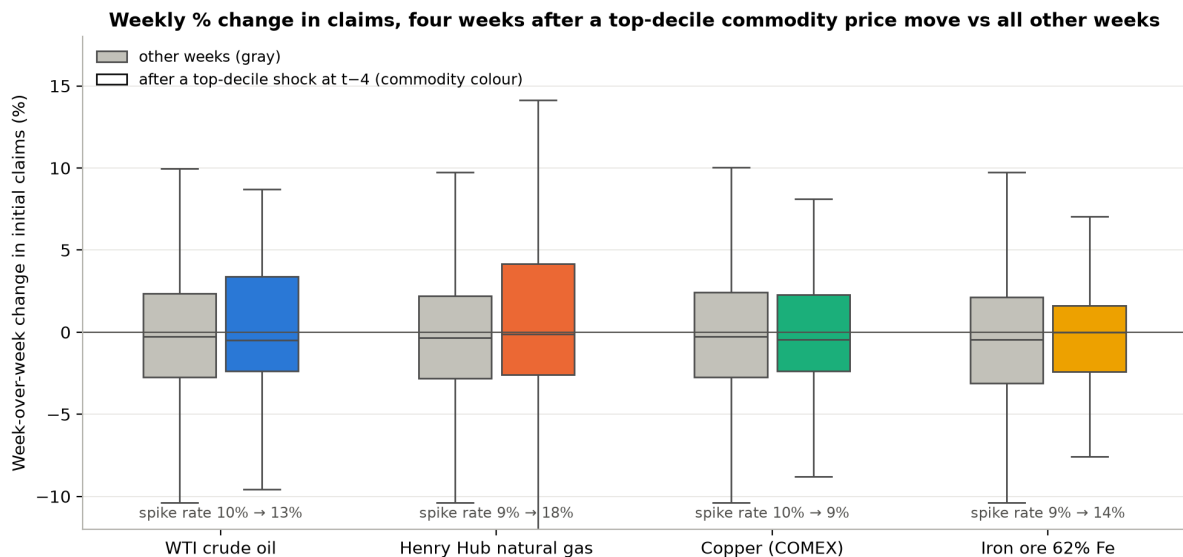


Figure 5. Distribution (median, interquartile box, $1.5 \times \text{IQR}$ whiskers; outliers beyond the axis hidden for legibility, not removed from the statistics) of the weekly % change in initial claims, split by whether the commodity recorded a top-decile absolute weekly price move four weeks earlier. Gray boxes: all other weeks; coloured boxes: weeks following a shock. Text under each pair gives the share of weeks that are claims spikes (rises above 5.2 %). Natural-gas shocks double the spike rate.

IV. Hypotheses and Analytical Plan

Notation: Y_t is the weekly change in log initial claims, $\Delta \log(\text{ICSA}_t)$; Δp_{t-k}^c is the % change in commodity c's weekly mean price k weeks earlier (from `add_lags()`); σ_t^c is its trailing 4-week volatility; R_t is the NBER recession flag; Z_t collects the controls (`usrec`, `covid_period`, lagged Y_t).

H1 — Energy shocks (crude oil). H_0 : $\beta^{\text{oil}}_k = 0$ for all $k \in \{0, \dots, 16\}$; H_1 : $\beta^{\text{oil}}_k \neq 0$ for at least one k , in $Y_t = \alpha + \sum_k \beta^{\text{oil}}_k \Delta p_{t-k}^{\text{oil}} + \gamma' Z_t + \varepsilon_t$. The EDA suggests $\beta^{\text{oil}}_{\{0,1\}} < 0$ (price collapses, not spikes, precede claims rises), so the test is two-sided.

H2 — Energy shocks (natural gas). H_0 : $\beta^{\text{gas}}_k = 0$ for all k ; H_1 : $\beta^{\text{gas}}_k > 0$ for some $k \in \{2, \dots, 8\}$. Motivated by the contingency result that a gas shock at $t-4$ doubles the spike probability.

H3 — Industrial-metal shocks (copper, iron ore). H_0 : $\beta^{\text{cu}}_k = \beta^{\text{fe}}_k = 0$ for all k ; H_1 : at least one differs from zero. Given the "Dr. Copper" pattern we expect negative coefficients (falling metal prices signal shrinking orders and later layoffs); the direction will be reported, not assumed.

H4 — Volatility, not only direction. H_0 : $\delta^c = 0$; H_1 : $\delta^c > 0$ in $Y_t = \alpha + \delta^c \sigma_{t-4}^c + \gamma' Z_t + \varepsilon_t$ for each commodity c : uncertainty about input costs raises layoffs independently of the sign of the price move.

H5 — Lag structure and regime dependence. H_0 : the lag k^* at which $|\beta^c_k|$ is maximal is the same in recession and expansion weeks and the interaction coefficient $\theta^c_k = 0$; H_1 : $\theta^c_k \neq 0$ in $Y_t = \alpha + \beta^c_k \Delta p_{t-k}^c + \theta^c_k \Delta p_{t-k}^c \cdot R_t + \lambda R_t + \gamma' Z_t + \varepsilon_t$ (transmission is faster and/or stronger in recessions). The proposal's expectation is $k^* \in [4, 12]$ weeks.

Table 13. Mapping of each hypothesis to columns of `master_weekly.csv`.

H	Target / response (Y)	Primary predictors (X)	Confounders / controls (Z)
H1	<code>icsa_pct_chg_1w</code> (alt.: $\Delta \log_{\text{icsa}}$; <code>icsa_spike</code> for classification, derived in the notebook)	<code>oil_pct_1w</code> (lag 0) + <code>oil_pct_lag1..16</code> from <code>add_lags()</code>	<code>usrec</code> , <code>covid_period</code> , lagged target (<code>icsa_pct_chg_1w</code> shifted 1–2), <code>gas_pct_lag*</code>
H2	<code>icsa_pct_chg_1w</code> / <code>icsa_spike</code>	<code>gas_pct_lag2..8</code>	<code>usrec</code> , <code>covid_period</code> , lagged target (<code>icsa_pct_chg_1w</code> shifted 1–2), <code>oil_pct_lag*</code>
H3	<code>icsa_pct_chg_1w</code> / <code>icsa_spike</code>	<code>copper_pct_1w</code> , <code>iron_pct_1w</code> + their <code>lag1..16</code> from <code>add_lags()</code>	<code>usrec</code> , <code>covid_period</code> , lagged target (<code>icsa_pct_chg_1w</code> shifted 1–2), <code>oil_pct_lag*</code>
H4	<code>icsa_pct_chg_1w</code>	<code>oil_vol_4w</code> , <code>gas_vol_4w</code> , <code>copper_vol_4w</code> , <code>iron_vol_4w</code> (each shifted 4 weeks)	<code>usrec</code> , <code>covid_period</code> , lagged target (<code>icsa_pct_chg_1w</code> shifted 1–2), own <code>{c}_pct_lag4</code>
H5	<code>icsa_pct_chg_1w</code>	<code>{c}_pct_lagk</code> $\times$ <code>usrec</code> (interaction) for the k^* found under H1–H3	<code>usrec</code> , <code>covid_period</code> , lagged target (<code>icsa_pct_chg_1w</code> shifted 1–2)

Analytical plan. H1–H3 will be tested with distributed-lag regressions on $\Delta \log$ claims using Newey–West (HAC) standard errors, complemented by Granger-causality F-tests; H4 adds the volatility terms; H5 adds the recession interaction and a rolling-window re-estimation. The predictive counterpart is a gradient-boosted regressor on the same features with time-series cross-validation, from which the strongest lags per commodity will be read off (permutation importance) and retained for a parsimonious final model. Scalars and the spike threshold will be refitted on the training window only. In week 14 the nine weeks of data generated during the semester will be pulled with the same scripts and used as a true out-of-sample test (trend matching and mean absolute error).

Reproducibility

Code and data: `src/collect.py` (raw pulls with manifest), `src/build_master.py` (weekly master dataset and data dictionary), `Milestone02_Analysis.ipynb` (all tables and figures in this draft, executed end-to-end with `jupyter nbconvert --execute`), `src/make_report.py` (this document). Environment: Python 3.13, pandas 3.0, statsmodels 0.15, yfinance 1.7, matplotlib 3.11, seaborn 0.13.

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